LAB 12 – Random Forest Implementation

2403A52006

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Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier, RandomForestClassifier

from sklearn.metrics import accuracy_score, precision_score, recall_score,
from sklearn.preprocessing import StandardScaler
```

Load Dataset

```
df = pd.read_csv("/content/UCI_Credit_Card.csv")

df.head()
```

	ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	.
0	1	20000.0	2	2	1	24	2	2	-1	-1	
1	2	120000.0	2	2	2	26	-1	2	0	0	
2	3	90000.0	2	2	2	34	0	0	0	0	
3	4	50000.0	2	2	1	37	0	0	0	0	
4	5	50000.0	1	2	1	57	-1	0	-1	0	

5 rows × 25 columns

Data Preprocessing

```
df = df.drop("ID", axis=1)
```

Define Features and Target

```
X = df.drop("default.payment.next.month", axis=1)
y = df["default.payment.next.month"]
```

Train/Test Split (80/20)

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

Feature Scaling

```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Train Decision Tree

```
dt_model = DecisionTreeClassifier(random_state=42)

dt_model.fit(X_train_scaled, y_train)

y_pred_dt = dt_model.predict(X_test_scaled)

acc_dt = accuracy_score(y_test, y_pred_dt)
prec_dt = precision_score(y_test, y_pred_dt)
rec_dt = recall_score(y_test, y_pred_dt)
roc_dt = roc_auc_score(y_test, y_pred_dt)

print("Decision Tree Results")
print("Accuracy:", round(acc_dt,4))
print("Precision:", round(prec_dt,4))
print("Recall:", round(rec_dt,4))
print("ROC-AUC:", round(roc_dt,4))
```

```
Decision Tree Results
Accuracy: 0.725
Precision: 0.3802
Recall: 0.4075
ROC-AUC: 0.6107
```

Train AdaBoost

```
ada_model = AdaBoostClassifier(
    n_estimators=50,
```

```
    random_state=42
)

ada_model.fit(X_train_scaled, y_train)

y_pred_ada = ada_model.predict(X_test_scaled)

acc_ada = accuracy_score(y_test, y_pred_ada)
prec_ada = precision_score(y_test, y_pred_ada)
rec_ada = recall_score(y_test, y_pred_ada)
roc_ada = roc_auc_score(y_test, y_pred_ada)

print("AdaBoost Results")
print("Accuracy:", round(acc_ada,4))
print("Precision:", round(prec_ada,4))
print("Recall:", round(rec_ada,4))
print("ROC-AUC:", round(roc_ada,4))
```

```
AdaBoost Results
Accuracy: 0.8167
Precision: 0.6778
Recall: 0.3092
ROC-AUC: 0.634
```

Train Random Forest

```
rf_results = []

estimators = [50, 100, 200]

for n in estimators:

    rf_model = RandomForestClassifier(
        n_estimators=n,
        max_depth=None,
        random_state=42
    )

    rf_model.fit(X_train_scaled, y_train)

    y_pred_rf = rf_model.predict(X_test_scaled)

    acc = accuracy_score(y_test, y_pred_rf)
    prec = precision_score(y_test, y_pred_rf)
    rec = recall_score(y_test, y_pred_rf)
    roc = roc_auc_score(y_test, y_pred_rf)

    rf_results.append([n, acc, prec, rec, roc])

rf_results
```

```
[[50,
  0.8158333333333333,
  0.6394101876675603,
```

```
  0.3632901751713633,
  np.float64(0.6529486933036249)],
 [100,
  0.816,
  0.6384105960264901,
  0.3670982482863671,
  np.float64(0.6544260176784941)],
 [200,
  0.8153333333333334,
  0.6354029062087186,
  0.36633663366336633,
  np.float64(0.6537251762300189)]]
```

Show Random Forest Results

```
rf_df = pd.DataFrame(
    rf_results,
    columns=["n_estimators","Accuracy","Precision","Recall","ROC-AUC"]
)

rf_df
```

	n_estimators	Accuracy	Precision	Recall	ROC-AUC
0	50	0.815833	0.639410	0.363290	0.652949
1	100	0.816000	0.638411	0.367098	0.654426
2	200	0.815333	0.635403	0.366337	0.653725

Next steps: Generate code with rf_df New interactive sheet

Compare All Models

```
print("\nModel Comparison")

print("\nDecision Tree:")
print("Accuracy:", round(acc_dt,4))
print("Precision:", round(prec_dt,4))
print("Recall:", round(rec_dt,4))
print("ROC-AUC:", round(roc_dt,4))

print("\nAdaBoost:")
print("Accuracy:", round(acc_ada,4))
print("Precision:", round(prec_ada,4))
print("Recall:", round(rec_ada,4))
print("ROC-AUC:", round(roc_ada,4))

print("\nRandom Forest Results:")
print(rf_df)
```

```
Model Comparison

Decision Tree:
Accuracy: 0.725
Precision: 0.3802
Recall: 0.4075
ROC-AUC: 0.6107

AdaBoost:
Accuracy: 0.8167
Precision: 0.6778
Recall: 0.3092
ROC-AUC: 0.634

Random Forest Results:
   n_estimators  Accuracy  Precision    Recall   ROC-AUC
0            50  0.815833   0.639410  0.363290  0.652949
1           100  0.816000   0.638411  0.367098  0.654426
2           200  0.815333   0.635403  0.366337  0.653725
```

Conclusion

Random Forest generally performs better than a single Decision Tree because it reduces variance by averaging multiple trees. Compared to AdaBoost, Random Forest can provide more stable performance for large datasets. The model with the highest ROC-AUC and accuracy is selected as the best-performing model